Dr. Pritam Mandal

Metallurgical & Materials Engineer

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Objective

Looking for opportunity in a Research & Development organization in Metallurgical & Materials Engineering.

Profile Summary

• Strong background in advanced materials design and surface engineering for aerospace and high-temperature applications.
• Research expertise in high-entropy alloys, Ni-superalloys, advanced coating technologies, and thermal barrier systems.
• Experienced in microstructure–property correlation via experimental characterization and machine-learning–based microstructure prediction.
• Expertise in Optical Microscopy with image analysis, SEM, XRD, tensile testing, micro-hardness testing, and corrosion behaviour.
• Certified Non-Destructive Testing (NDT) Level-II professional (VT, UT, RT, DPT).
• Experience in AI data annotation, scientific reasoning evaluation, and research-quality problem design for frontier LLMs in computational materials science.

Skills

Data-based modelling (Machine Learning, Python) • LLM model training (STEM experts) • CALPHAD • Material characterization (SEM, EDS, XRD, TEM, DSC/TGA, mechanical, electrochemical, thermal) • NDT Level II (VT, UT, RT, DPT) • Advanced surface protection & coatings • Metallurgical process simulations

Experience

Research Associate (RA)

SEST, University of Hyderabad · Apr 2026 – Present

NMDC-sponsored: Utilization of BHQ Baladilla Region for the Production of Ferrosilicon

• Designing experiments on phase transformations and metallurgical processing relevant to ferrosilicon production.
• Performing microstructural characterization and data-driven analysis using Python to generate process insights.
• Contributing to multi-disciplinary R&D discussions; supporting lab management, resource planning, and safety compliance.

Research Quality Specialist – Materials Science (Contract)

Turing, Remote · Dec 2025 – Apr 2026

• Developed and validated research-grade computational materials science problems for AI model training and evaluation.
• Designed Python-based scientific solutions and assessed technical accuracy, logical reasoning, and solution quality.
• Reviewed AI-generated responses and collaborated with global experts to improve high-quality STEM datasets for frontier LLMs.

Research Areas

Thermal Barrier Coating • High-Entropy Alloy/Oxides • Ni-based Superalloys • Aeroengine Materials • Advanced Engineering Materials • Data-Based Process Modelling • Iron-Making Process Simulations & Computational Metallurgy

Education

Ph.D. in Metallurgy and Materials Engineering — Completed 19/08/2026

IIEST, Shibpur

M.Tech. in Materials Engineering — 07/2020

IIEST, Shibpur

B.Tech. in Mechanical Engineering — 07/2017

Maulana Abul Kalam Azad University of Technology (formerly WBUT)

Publications (Total 7)

Google Scholar: scholar.google.com/citations?user=g0f7HBMAAAAJ | ORCID: 0000-0003-1924-5027

1. Mandal, P., Choudhury, A., Mallick, A.B., Ghosh, M. Phase Prediction in High Entropy Alloys by Various Machine Learning Modules Using Thermodynamic and Configurational Parameters. Met. Mater. Int. 29, 38–52 (2023).

2. Mandal, P., Choudhury, A., Mallick, A.B., Ghosh, M. A voting-based ensemble classifier to predict phases and crystal structures of high entropy alloys. J. Alloys Metall. Syst. (2023).
3. Adhikari, J., Saha, P., Mandal, P. et al. A Review on High Entropy Alloys as Metallic Biomaterials. Biomedical Materials & Devices (2025).
4. Dey, D., Pal, A., Biyani, P. et al. Developing new high-entropy alloys with enhanced hardness using a hybrid machine learning approach. J Mater Sci 60, 4820–4845 (2025).
5. Dey, D., Das, S., Pal, A. et al. Improved machine learning framework for prediction of phases and crystal structures of high entropy alloys. J. Alloys Metall. Syst. 9, 100144 (2025).
6. Pal, A., Verma, R., Mandal, P. et al. Influence of Alloying Elements and Annealing Temperature on Lankford Constant of High Strength Dual Phase Steels. J. Inst. Eng. India Ser. D (2026).
7. Roy, S. et al. Utilizing Machine Learning Algorithms to Predict Accuracy of the Index of Relative Tectonic Activity (IRTA). IEEE Access 12, 60482–60495 (2024).

Referees

Dr. Manojit Ghosh – Ph.D. Supervisor, Professor, Metallurgy & Materials Engg., IIEST Shibpur
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